

Module 14: Deadlock Avoidance & Banker's Algorithm

1. Safe State & Banker's Trace

State Spaces Relationship

All Operating System States

UNSAFE STATE (Deadlock Possible)

DEADLOCK

SAFE STATE (Guaranteed Safe Sequence Exists)

Deadlock Avoidance keeps system strictly in SAFE STATE!

Banker's Algorithm Step-by-Step Resolution Trace

Available Vector: [3, 3, 2] | Need = Max - Allocation
Test Condition for Process P_i: Need_j ≤ Work (Available)

Step 1: Check P2: Need[1,2,2] ≤ Avail[3,3,2] ⇒ SATISFIED! New Avail = [5, 3, 2]

Step 2: Check P4: Need[0,1,1] ≤ Avail[5,3,2] ⇒ SATISFIED! New Avail = [7, 4, 3]

Step 3: Check P5: Need[4,3,1] ≤ Avail[7,4,3] ⇒ SATISFIED! New Avail = [7, 4, 5]

Step 4: Check P1: Need[7,4,3] ≤ Avail[7,4,5] ⇒ SATISFIED! New Avail = [7, 5, 5]

Step 5: Check P3: Need[6,0,0] ≤ Avail[7,5,5] ⇒ SATISFIED! New Avail = [10, 5, 7]

SAFE SEQUENCE: < P2, P4, P5, P1, P3 > (System is SAFE!)

2. Golden Formula & Safe Sequence

Need Matrix Formula:
$$\text{Need}[i][j] = \text{Max}[i][j] - \text{Allocation}[i][j]$$

Safe Sequence Found: < P2, P4, P5, P1, P3 >